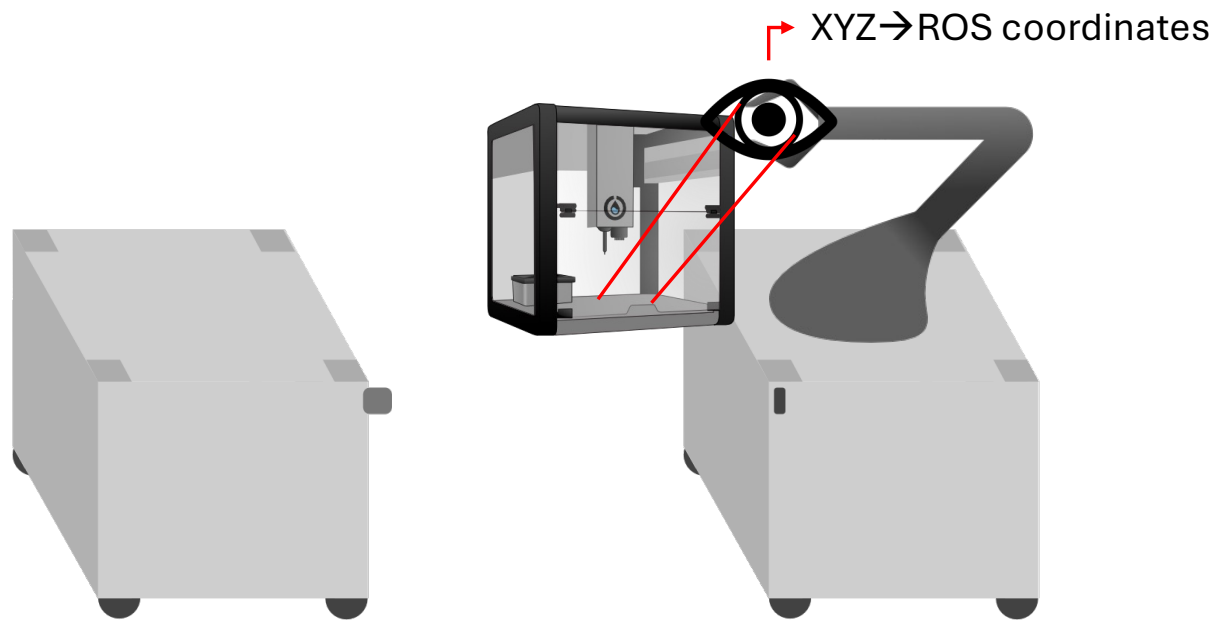


Robotic arm autocalibration

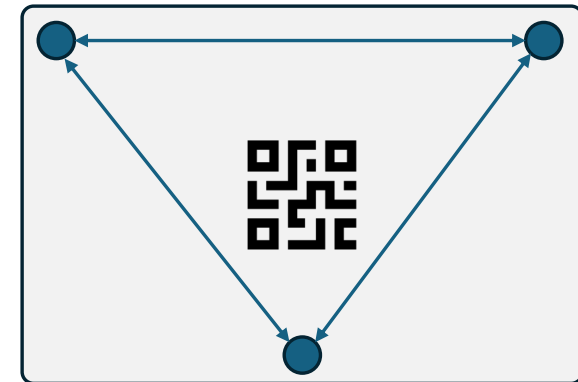
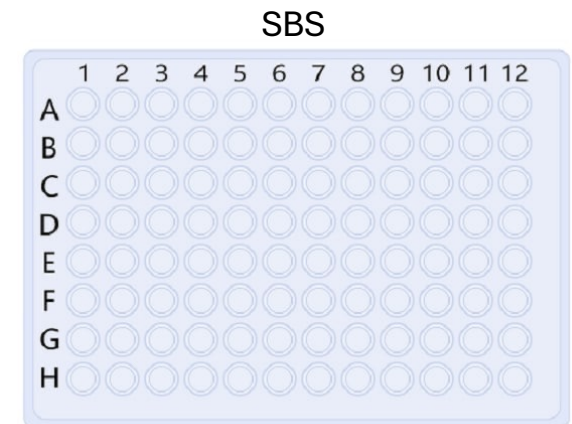
A step towards simplifying laboratory expansion





The goal of the project

- The need:
 - We have a robotic arm
 - We have a modular workbench
 - We need a code-free calibration method
- Building an autonomous calibration system
 - Based on camera information
 - Using our open source robotic arm
 - Using our modular workbenches
- Current project idea
 - Create a 3d-printed autocalibrator in SBS-format



Step 1: find plate (QR)

Step 2: Triangulation (circles)



The skills required



Image recognition



Translating
triangulation into ROS
coordinates



Implementing a
camera into the setup



Writing the code for
detecting positions



Current stage

- Robotic arm ROS implementation is getting developed
- Workbench is in stage 1 of Development
- Summer: They have been assembled



The need

Reproducibility remains a significant challenge in biotech. Even with detailed experimental descriptions, numerous free parameters make it difficult to replicate experiments. To address this, we are developing a laboratory that enables fully downloadable protocols and standardized laboratory setups.



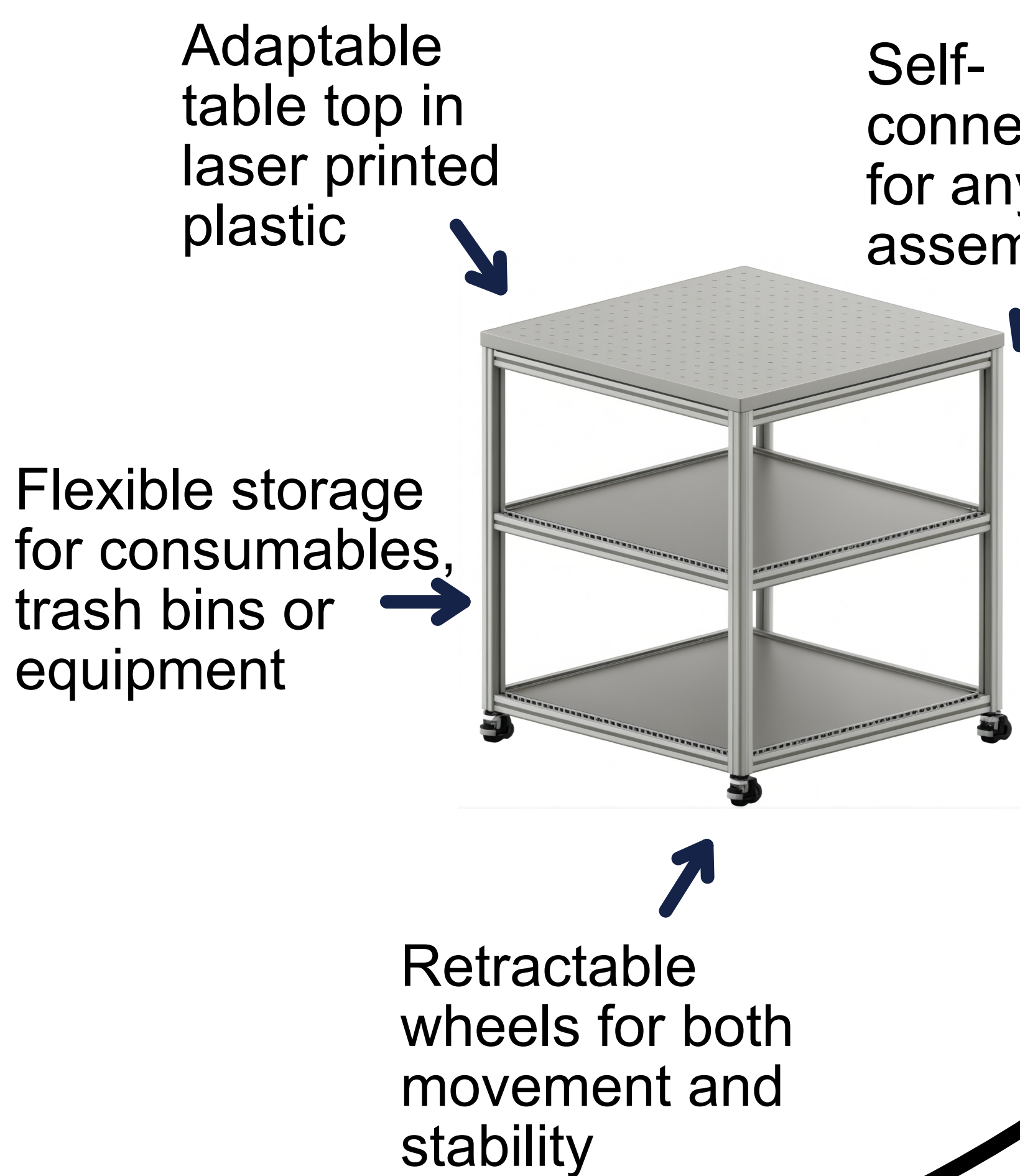
World Wide Lab

The solution

For automated lab setups to be ‘translatable,’ two key challenges must be addressed. First, robots must be easily reconfigurable to ensure consistency across labs. We tackle this with universal, open-source worktables and robotic arms, complete with CAD models and guidelines. Second, software must be streamlined. Using PyLabRobot, SILA, and ROS, we aim to create a unified, GitHub-friendly language for lab automation.

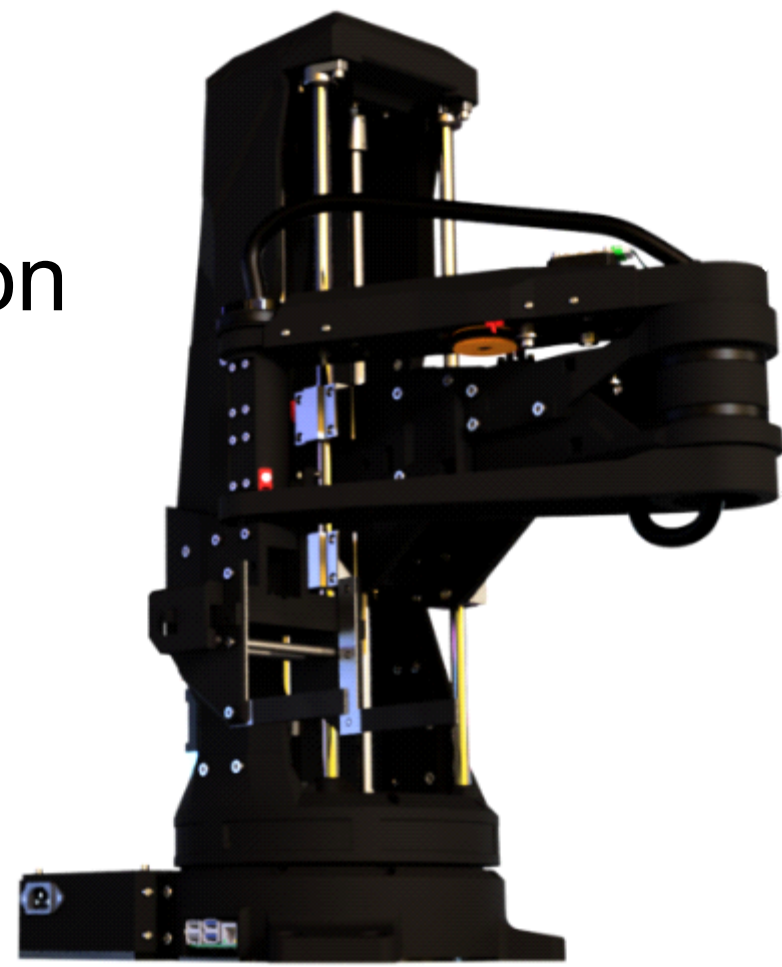
Thanks to: Emil Hemmingsen, Mads Bjørnskov, William Sommer

Thanks to: Sebastian Storz, August Løngren Iversen, Sebastian Nielsen



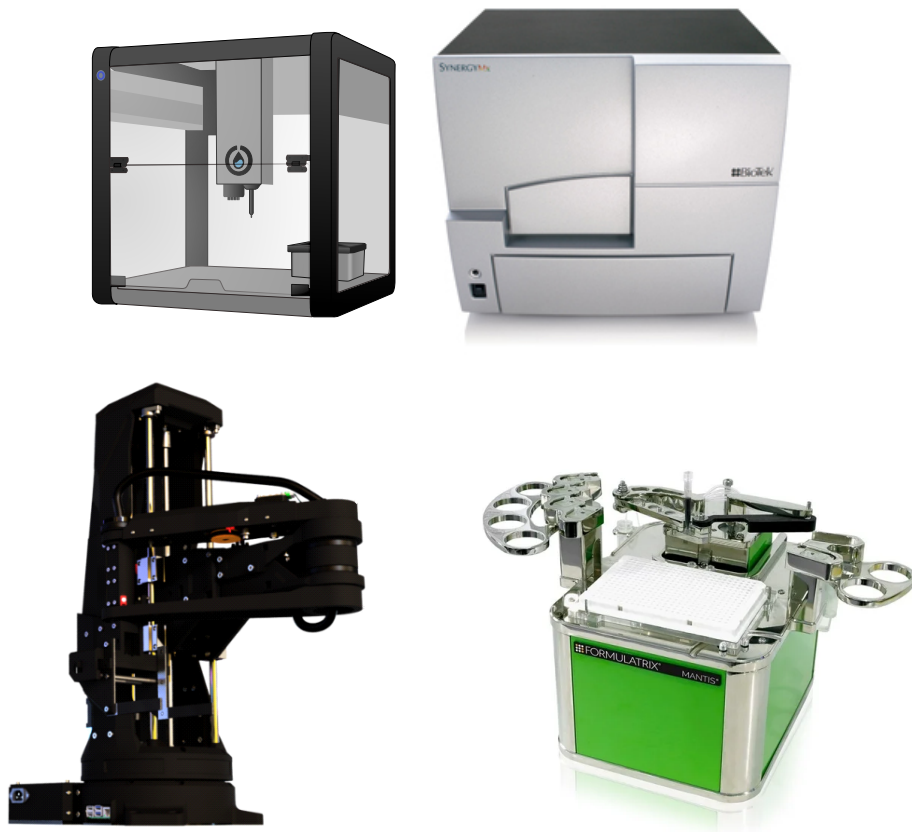
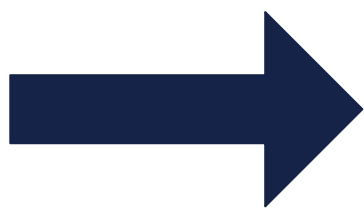
Open source robotic arm:

- Lower entry barrier for automation
- Modifiable
- ROS compatible



NFC

SILA Rapid Integration
Software input-output normalization improves connectivity



Downloadable protocols enable exact copied experiments

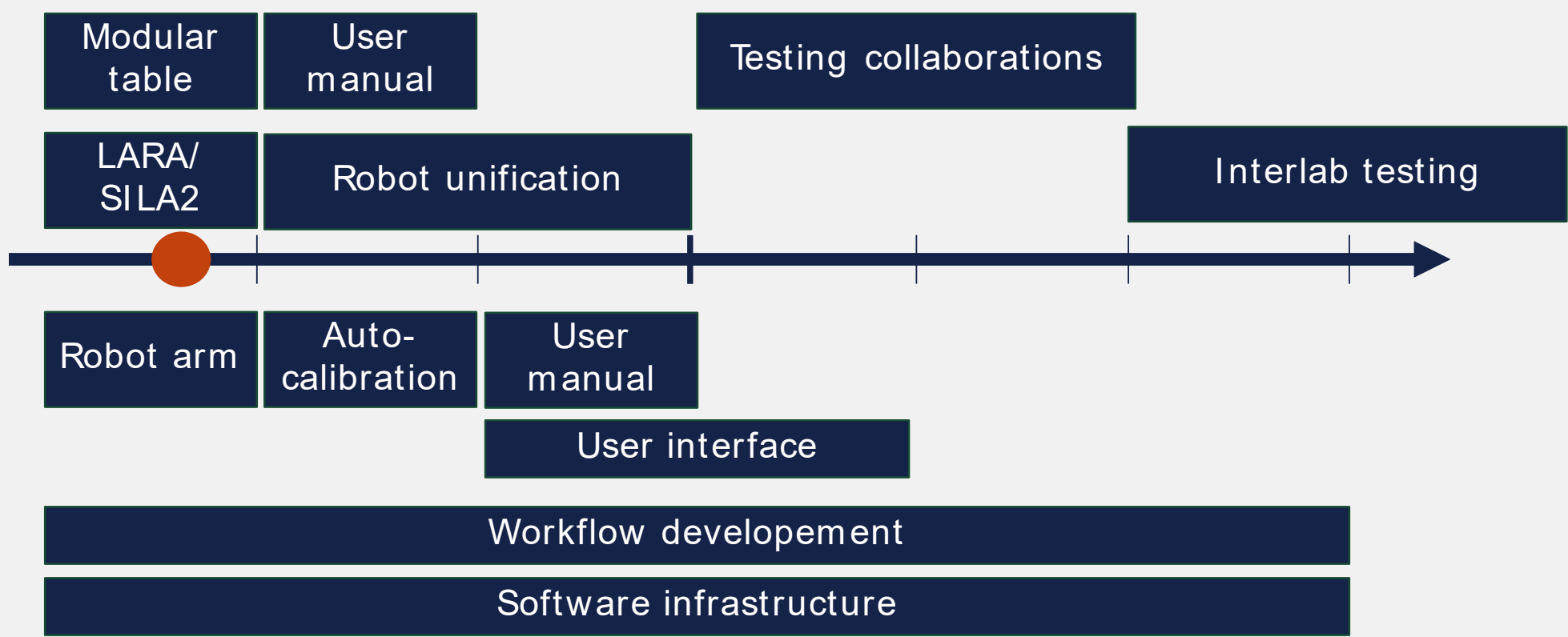


World wide parameter-locked experiments

Thanks to: Florin-Bogdan Gavriluta, Kevin Skytte, Haoyue Hu and Shuhui Li

Current stage

We started the practical experiments in January 2025 and hope to have the first prototypes this summer. However, we expect longer times before the system becomes user friendly.



Next stage

Next semester (autumn 2025) we hope to initiate an autocalibration system for the robotic arm, simplifying the assembly of robots. We also hope to implement the plate-tracking and continuing the driver development.

